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| <b>Ex.No:</b> |  |
| <b>Date:</b>  |  |

**AIM:**

**ALGORITHM:**

**CODE:**

```
from Crypto.Cipher import DES
from Crypto.Util.Padding import pad, unpad
key = b'12345678'
cipher = DES.new(key, DES.MODE_ECB)
plaintext = input("Enter Plaintext: ")
ciphertext = cipher.encrypt(pad(plaintext.encode(), DES.block_size))
```

```
print("Encrypted Data:", ciphertext.hex())  
  
decipher = DES.new(key, DES.MODE_ECB)  
  
decrypted = unpad(decipher.decrypt(ciphertext), DES.block_size)  
  
print("Decrypted Data:", decrypted.decode())
```

**OUTPUT:**

```
Enter Plaintext: HELLO  
Encrypted Data: 8b6a14412de58903  
Decrypted Data: HELLO
```

**RESULT:**

|               |  |
|---------------|--|
| <b>Ex.No:</b> |  |
| <b>Date:</b>  |  |

**AIM:**

**ALGORITHM:**

**CODE:**

```
def gcd(a, b):  
    while b:  
        a, b = b, a % b  
    return a  
  
def mod_inverse(e, phi):  
    for d in range(1, phi):  
        if (e * d) % phi == 1:
```

```
        return d

p = 17
q = 11
n = p * q
phi = (p - 1) * (q - 1)
e = 7
d = mod_inverse(e, phi)
print("Public Key:", (e, n))
print("Private Key:", (d, n))
message = int(input("Enter Message (<187): "))
cipher = pow(message, e, n)
print("Encrypted Message:", cipher)
plain = pow(cipher, d, n)
print("Decrypted Message:", plain)
```

#### **OUTPUT:**

```
Public Key: (7, 187)
Private Key: (23, 187)
Enter Message (<187): 11
Encrypted Message: 88
Decrypted Message: 11
```

#### **RESULT:**

|               |  |
|---------------|--|
| <b>Ex.No:</b> |  |
| <b>Date:</b>  |  |

**AIM:**

**ALGORITHM:**

231401102

**CODE:**

```
import hashlib  
message = input("Enter Message: ")  
hash_value = hashlib.md5(message.encode())  
print("MD5 Digest:", hash_value.hexdigest())
```

**OUTPUT:**

```
Enter Message: Cryptology  
MD5 Digest: c6ae9d0a089311bab119969161ca5cce
```

**RESULT:**

|               |  |
|---------------|--|
| <b>Ex.No:</b> |  |
| <b>Date:</b>  |  |

**AIM:**

**ALGORITHM:**

231401102

**CODE:**

```
import hashlib  
  
message = input("Enter Message: ")  
  
hash_value = hashlib.sha1(message.encode())  
  
print("SHA-1 Digest:", hash_value.hexdigest())
```

**OUTPUT:**

---

```
Enter Message: Cryptology  
SHA-1 Digest: ba7160026c971c61eca61f590d68b3de9fadc0f3
```

**RESULT:**